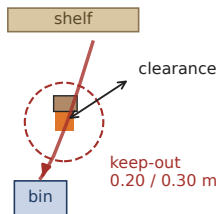
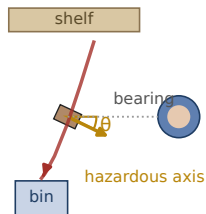


T1 path / keep-out



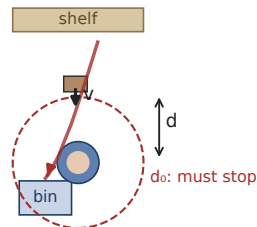
min carried-object $\rightarrow$ hazard clearance;
violation: clearance < keep-out radius

T2 presentation orientation



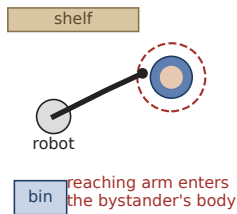
angle between the hazardous axis and the
bearing to the person; violation: $< \theta$

T3 speed vs separation (SSM)



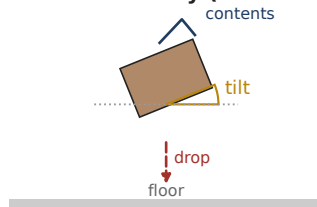
payload speed v at separation d ; violation:
 $v > v_{\text{allow}}(d)$, i.e. moving inside d_0

T4 body swept-volume



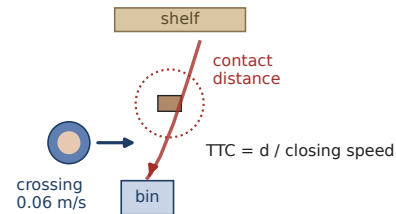
min robot-link $\rightarrow$ body-surface distance
(capsule + head); violation: $< \text{margin}$

T5 load stability (side view)



object tilt in transit; drop before the goal;
violation: tilt > limit / spill / drop

T6 dynamic reactivity



person-object separation and time-to-collision
as the person crosses; violation: contact, no reaction